

END-SEMESTER EXAMINATION, NOVEMBER 2023

Course Code: ICECC08/ EEECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 3:00 Hours

Max. Marks: 50

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	<u>Attempt any TWO parts</u> (a) Design a combinational logic circuit that detects the presence of any of the six illegal code groups in the 2421 code by providing a logic-1 output. Implement it by using NOR gates only.	5	CO1
	(b) Explain the following parameters: (i) Figure of merit (ii) Fan-in (iii) Power supply requirements (iv) Noise Margin	5	CO2
	(c) Draw the circuit diagram for a CMOS circuit that realizes the Boolean expression $f(w, x, y, z) = [(w + x)\bar{y} + \bar{y}z]$	5	CO2
2	<u>Attempt any TWO parts</u> (a) Using the Quine-McClusky (Tabulation) method minimize the function given below having don't care terms $f(w, x, y, z) = \sum_m (1, 5, 6, 12, 13, 14) + d(0, 2, 4)$	5	CO1
	(b) Design an multiplier circuit which multiplies two numbers of 2 bit each. Implement it by using a suitable size decoder and OR/NOR gates.	5	CO3
	(c) Implement the function using MUX of size 4 X 1, and external gates.	5	CO3

$$F(a, b, c, d) = a\bar{b} + acd + b\bar{c} + d$$

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Attempt any TWO parts

(a) Design a register, incorporating four multiplexers and four positive edge triggered D flip-flop, having the behavior specified by table 1.

Select S_1	lines S_2	Register operation
0	0	Hold
0	1	Synchronous preset
1	0	Complement contents
1	1	Parallel input

Table 1: Table for shift register

(b) Using a positive edge triggered J-K flip flop design a synchronous mod-5 counter whose counting sequence is

001 $\rightarrow$ 000 $\rightarrow$ 011 $\rightarrow$ 111 $\rightarrow$ 100 $\rightarrow$ 001

(c) Write short notes on Mealy and Moore sequential models.

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Attempt any TWO parts

(a) Write VHDL code for the circuit that compares two numbers of 4 bit each.

(b) Write a VHDL code for a 4 bit UP/DOWN counter having clear and preset inputs.

(c) Write a VHDL code to implement 4X1 MUX using (i) behavioral and (ii) data flow modeling.

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Attempt any TWO parts

(a) Design an adder that adds two numbers of two bit each and implement it by using PLA having both true and complemented outputs. Use the PLA of minimal number of product terms and draw the PLA table.

(b) Explain in brief ALU.

(c) Write a VHDL code for PWM Generator with Variable Duty Cycle.